THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two dice are thrown simultaneously. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{31}{36}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{5}{36}$

2. A coin is tossed three times. The probability of getting exactly two heads is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{3}{8}$

3. A letter is chosen at random from the word 'MATHEMATICS'. The probability that it is a vowel is: 1

(a) $\frac{7}{11}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{4}{11}$

4. A bag contains 4 red and 6 blue balls. 2 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{6}{13}$ **(c)** $\frac{7}{12}$ **(d)** $\frac{5}{12}$

5. A box contains 20 tickets numbered 1 to 20. One ticket is drawn at random. The probability that the number drawn is a multiple of 3 is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{3}{10}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{7}{20}$

6. A lot of 135 ball pens contains 15 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{119}{135}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{121}{135}$ **(d)** $\frac{8}{9}$

7. Two dice are thrown simultaneously. The probability that the sum of the numbers is 7 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{7}{36}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{1}{6}$

8. A coin is tossed three times. The probability of getting at least two heads is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{4}{9}$

9. A letter is chosen at random from the word 'PROBABILITY'. The probability that it is the letter B is: 1

(a) $\frac{1}{11}$ **(b)** $\frac{9}{11}$ **(c)** $\frac{2}{11}$ **(d)** $\frac{3}{11}$

10. A bag contains 5 red and 5 blue balls. 5 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{11}{15}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{1}{3}$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A coin is tossed three times. Find the probability of getting at least one tail. 2

12. A red die and a blue die are rolled together. Find the probability that the sum is less than 5. 2

13. A letter is chosen at random from the word 'EDUCATION'. Find the probability that it is a vowel. 2

14. A coin is tossed three times. Find the probability of getting a head on the first toss. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two dice are thrown simultaneously. Find the probability that the product is a perfect square. 3

16. A box contains 18 tickets numbered 1 to 18. One ticket is drawn at random. Find the probability that the number drawn is a multiple of 3. 3

17. A bag contains 3 red and 9 blue balls. 3 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 3

OR

Two dice are thrown simultaneously. Find the probability that the sum is a multiple of 4. 3

18. A box contains 28 tickets numbered 1 to 28. One ticket is drawn at random. Find the probability that the number drawn is a prime number. 3

OR

A bag contains 7 red and 5 blue balls. 5 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 8,
(ii) both dice show the same number (a doublet),
(iii) the product of the numbers is 12.

OR

A bag contains 4 red and 8 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. Find the value of x. 5

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 7,
(ii) the sum is at least 10,
(iii) the sum is less than 5.

OR

A bag contains 5 red and 7 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{8}{15}$. Find the value of x. 5

END OF QUESTION PAPER — Total: 40 Marks